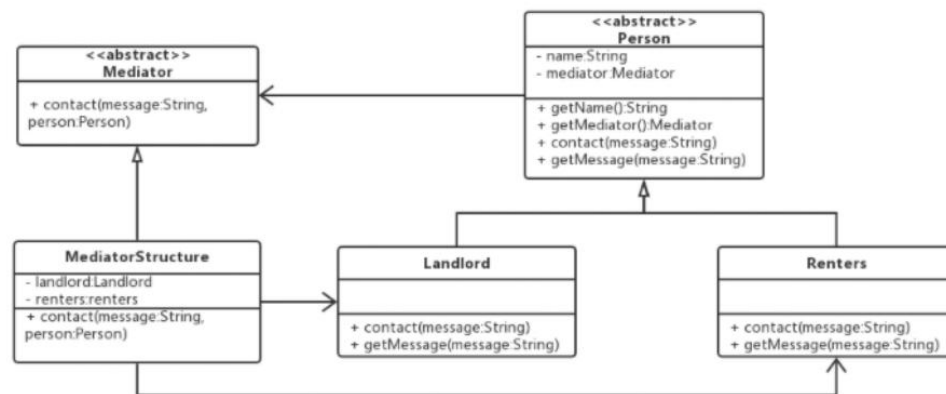


2. Use the mediator pattern to simulate a rental scenario. Write a Test class to test the Result. The following figure is the scenario UML diagram.



Landlord

```

package experiment21_2;

class Landlord extends Person{

    public Landlord(String name, Mediator mediator){
        super(name, mediator);
    }

    @Override
    public void contact(String message) {
        System.out.println("Landlord " + name + " sends
message: " + message);
        mediator.contact(message, this);
    }
}
  
```

```

@Override

public void getMessage(String message) {

    System.out.println("Landlord " + name + " gets
message: " + message);

}

}

```

Mediator

```

package experiment21_2;

abstract class Mediator {

    public abstract void contact(String message, Person
person);

}

```

MediatorStructure

```

package experiment21_2;

class MediatorStructure extends Mediator{

    private Landlord landlord;

    private Renters renters;

    public void setLandlord(Landlord landlord) {

        this.landlord = landlord;

    }

}

```

```

    public void setRenters(Renters renters) {
        this.renters = renters;
    }

    @Override
    public void contact(String message, Person person) {
        if(person == landlord){
            renters.getMessage(message);
        }else if(person == renters){
            landlord.getMessage(message);
        }
    }
}

```

Person

```

package experiment21_2;

abstract class Person {
    protected String name;
    protected Mediator mediator;
    public Person(String name, Mediator mediator){
        this.name = name;
    }
}

```

```

        this.mediator = mediator;
    }

    public String getName(){
        return name;
    }

    public Mediator getMediator(){
        return mediator;
    }

    public abstract void contact(String message);
    public abstract void getMessage(String message);
}

```

Renters

```

package experiment21_2;

class Renters extends Person{
    public Renters(String name, Mediator mediator){
        super(name, mediator);
    }

    @Override
    public void contact(String message) {
        System.out.println("Renter " + name + " sends

```

```

message: " + message);

    mediator.contact(message, this);

}

@Override

public void getMessage(String message) {

    System.out.println("Renter " + name + " gets message:
" + message);

}

}

```

Test

```

package experiment21_2;

public class Test {

    public static void main(String[] args) {

        MediatorStructure mediatorStructure = new
MediatorStructure();

        Landlord landlord = new Landlord("Lucy",
mediatorStructure);

        Renters renter = new Renters("Jack",
mediatorStructure);

        mediatorStructure.setLandlord(landlord);
    }
}

```

```
mediatorStructure.setRenters(renter);  
renter.contact("I want to rent a house.");  
landlord.contact("Ok, 2000 yuan per month");  
}  
}
```

Output

```
D:\JDK\bin\java.exe "-javaagent:D:\idea\IntelliJ IDEA 2021...  
Renter Jack sends message: I want to rent a house.  
Landlord Lucy gets message: I want to rent a house.  
Landlord Lucy sends message: Ok, 2000 yuan per month  
Renter Jack gets message: Ok, 2000 yuan per month  
  
Process finished with exit code 0
```